🏛️ Government E-Governance Service Management System (EGSMS)

Language: ANSI C
Architecture: Structured Enterprise Console System
Storage: Binary File-Based Persistence
Version: 1.0

📌 Executive Summary

The Government E-Governance Service Management System (EGSMS) is a structured public administration simulation platform developed entirely in ANSI C.

The system integrates core governance components including:

• Secure administrator authentication
• Citizen service request submission
• Request status processing
• Digital tax calculation
• Revenue analytics
• Timestamp-based audit logging

EGSMS models centralized digital governance workflow using deterministic structured programming and binary file persistence.

🏗️ System Architecture

EGSMS follows a layered logical architecture:

• Authentication Layer
• Citizen Service Request Module
• Status Processing Module
• Tax Calculation Engine
• Revenue Analytics Engine
• Audit Logging Layer
• Binary Persistence Layer

Logical separation of functional responsibilities is maintained within a single source file.

🔐 Authentication Module

The system ensures secure administrative access.

Features:

• Automatic admin initialization on first execution
• Password hashing using djb2 algorithm
• No plaintext credential storage
• Binary verification against `egov_users.dat`

Default Credentials:

Username: admin
Password: admin123

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

🏛️ Citizen Service Request Module

Each service request contains:

• Citizen ID
• Name
• Address
• Service Type
• Status

Default Status:

Pending

Functional Capabilities:

• Submit service request
• View service requests
• Update status (Approved / Rejected)

Storage File: `service_requests.dat`

🔄 Status Processing Module

The system allows administrative status updates.

Operational Flow:

• Locate citizen request
• Modify status field
• Rewrite record using file pointer repositioning

This simulates digital approval workflow.

💰 Tax Calculation Engine

Each tax record contains:

• Tax Record ID
• Citizen ID
• Annual Income
• Tax Amount

Tax Formula:

```
Tax Amount = Income × 15%
```

Functional Capabilities:

• Calculate tax
• Persist tax record

Storage File: `tax_records.dat`

📊 Revenue Analytics Engine

The analytics module processes tax records and computes:

• Mean (Average Tax)
• Variance
• Standard Deviation

Formulas Used:

```
Mean = Σ Tax / N
Variance = Σ (Tax − Mean)² / N
Standard Deviation = √Variance
```

Purpose:

• Analyze public revenue distribution
• Measure taxation variability
• Evaluate governance financial patterns

📝 Audit Logging

All major system activities are recorded in:

`egov_logs.txt`

Logged Events:

• Admin login
• Service submission
• Status update
• Tax calculation
• Analytics execution
• System exit

Each log entry includes timestamp metadata.

💾 Data Storage Design

Binary files used:

File	Purpose
egov_users.dat	Authentication records
service_requests.dat	Citizen service requests
tax_records.dat	Tax database
egov_logs.txt	Audit logs

Binary storage ensures:

• Structured record alignment
• Efficient data persistence
• Faster read/write operations
• Secure governance data management

⚙️ Compilation & Execution

Compile:

```
gcc egovernance_system.c -o egov -lm
```

Run:

```
./egov
```

🧠 Engineering Highlights

• ANSI C structured programming
• Public service workflow abstraction
• File-based governance database simulation
• Tax computation logic
• Statistical revenue analysis
• Deterministic memory handling
• Timestamp-based accountability tracking

🎯 Project Significance

EGSMS demonstrates the ability to design and implement a centralized digital governance workflow using low-level systems programming.

It reflects:

• Administrative process modeling
• Structured data persistence
• Revenue analytics integration
• Secure authentication design
• Enterprise-grade service management logic

🏁 Conclusion

The Government E-Governance Service Management System (EGSMS) is a complete public administration simulation platform implemented in pure ANSI C.

It integrates:

Authentication
Service Request Management
Status Processing
Tax Calculation
Revenue Analytics
Audit Logging

All within a structured and deterministic runtime architecture.